BT101
Introductory Biology
(Lecture 7)

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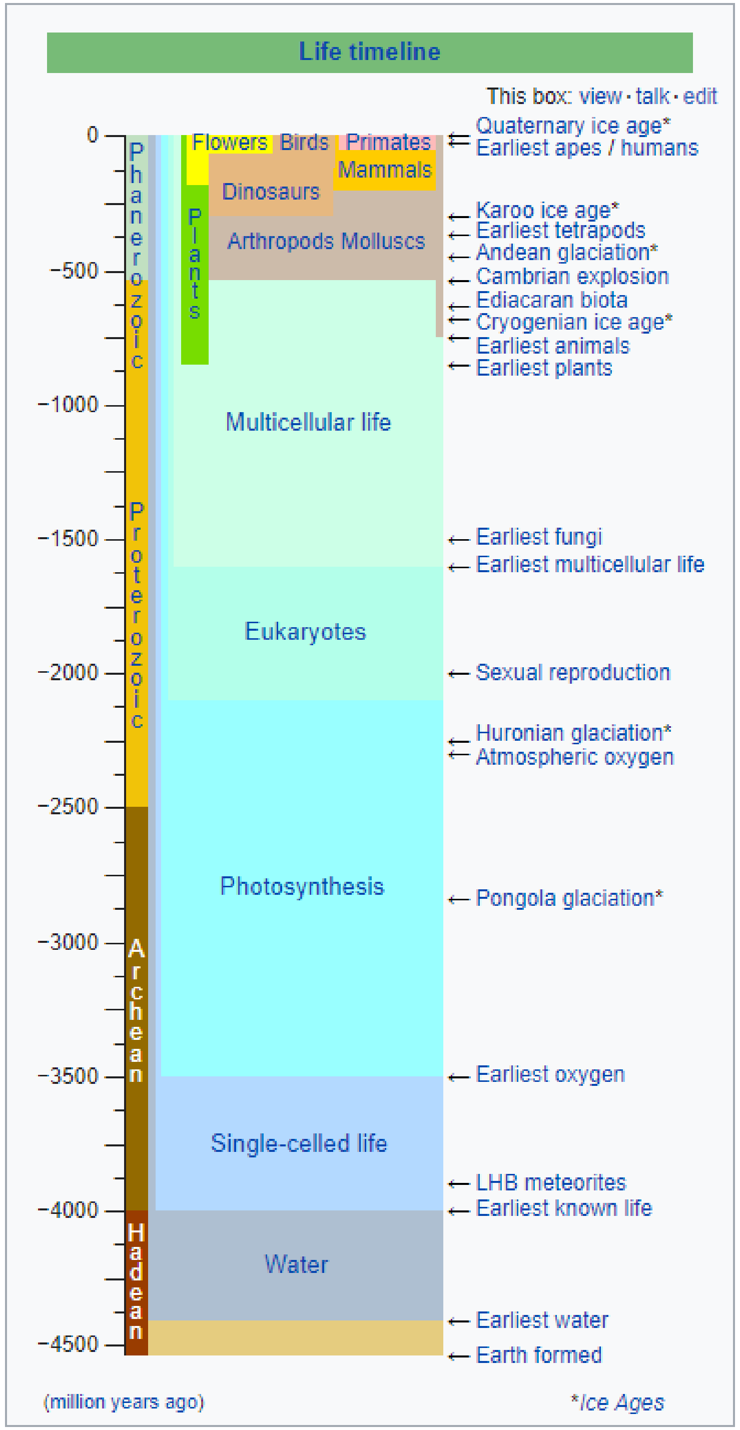
Introduction

Evolution of life:

- ❑ Origin of Life;
- ❑ Darwin's concepts of evolution
- ❑ Biodiversity

Cell, the structural and functional unit of life:

- ❑ **Three domains of life**
- ❑ Cell types, cell organelles and structure
- ❑ Basic biomolecules of cell


Life timeline
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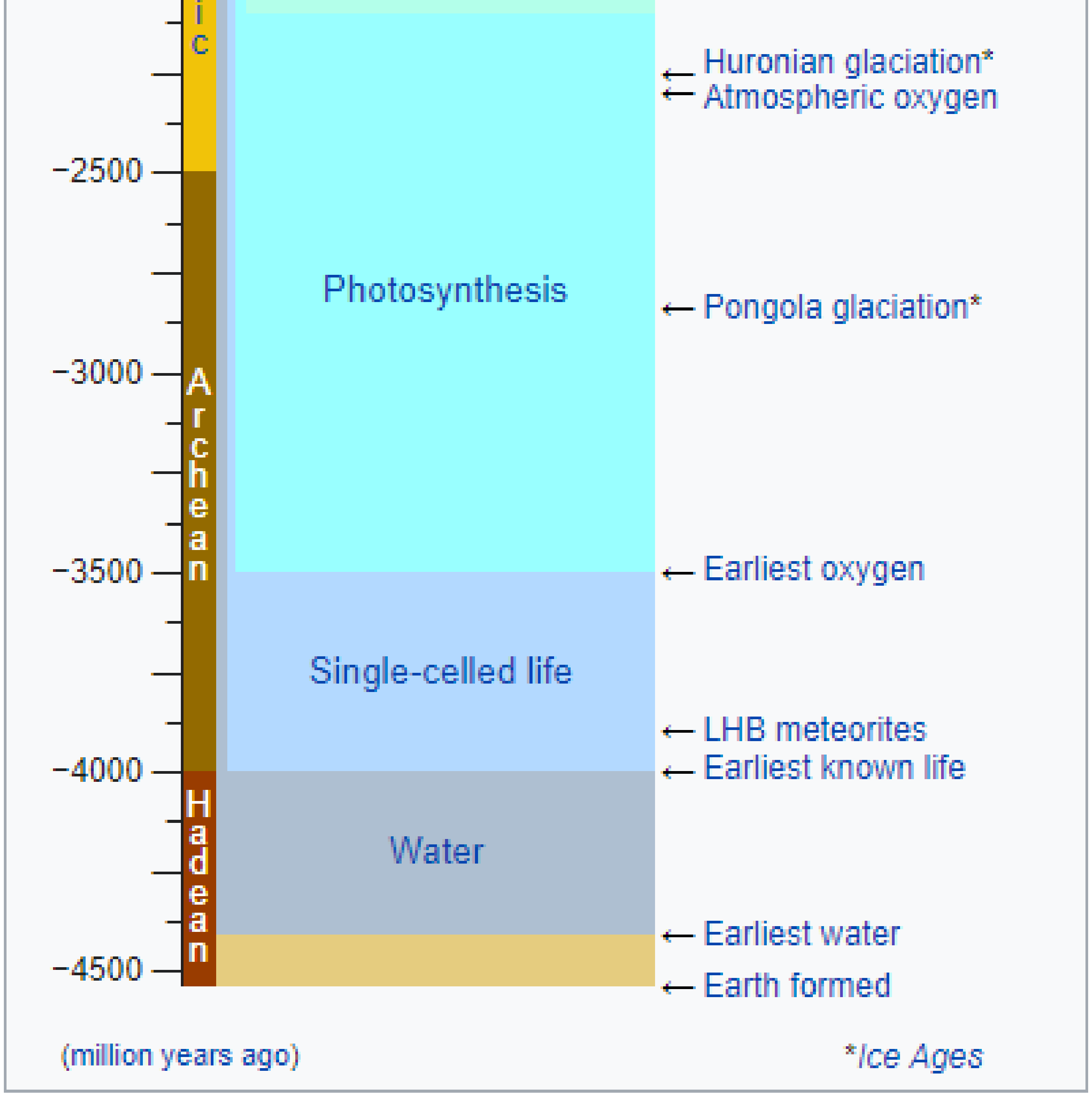
Phanerozoic
Proterozoic
Archean
Hadean
Flowers
Birds
Primates
Mammals
Dinosaurs
Plants
Arthropods Molluscs
Multicellular life
Eukaryotes
Photosynthesis
Single-celled life
Water
← Quaternary ice age*
← Earliest apes / humans
← Karoo ice age*
← Earliest tetrapods
← Andean glaciation*
← Cambrian explosion
← Ediacaran biota
← Cryogenian ice age*
← Earliest animals
← Earliest plants
← Earliest fungi
← Earliest multicellular life
← Sexual reproduction
← Huronian glaciation*
← Atmospheric oxygen
← Pongola glaciation*
← Earliest oxygen
← LHB meteorites
← Earliest known life
← Earliest water
← Earth formed
(million years ago)
*Ice Ages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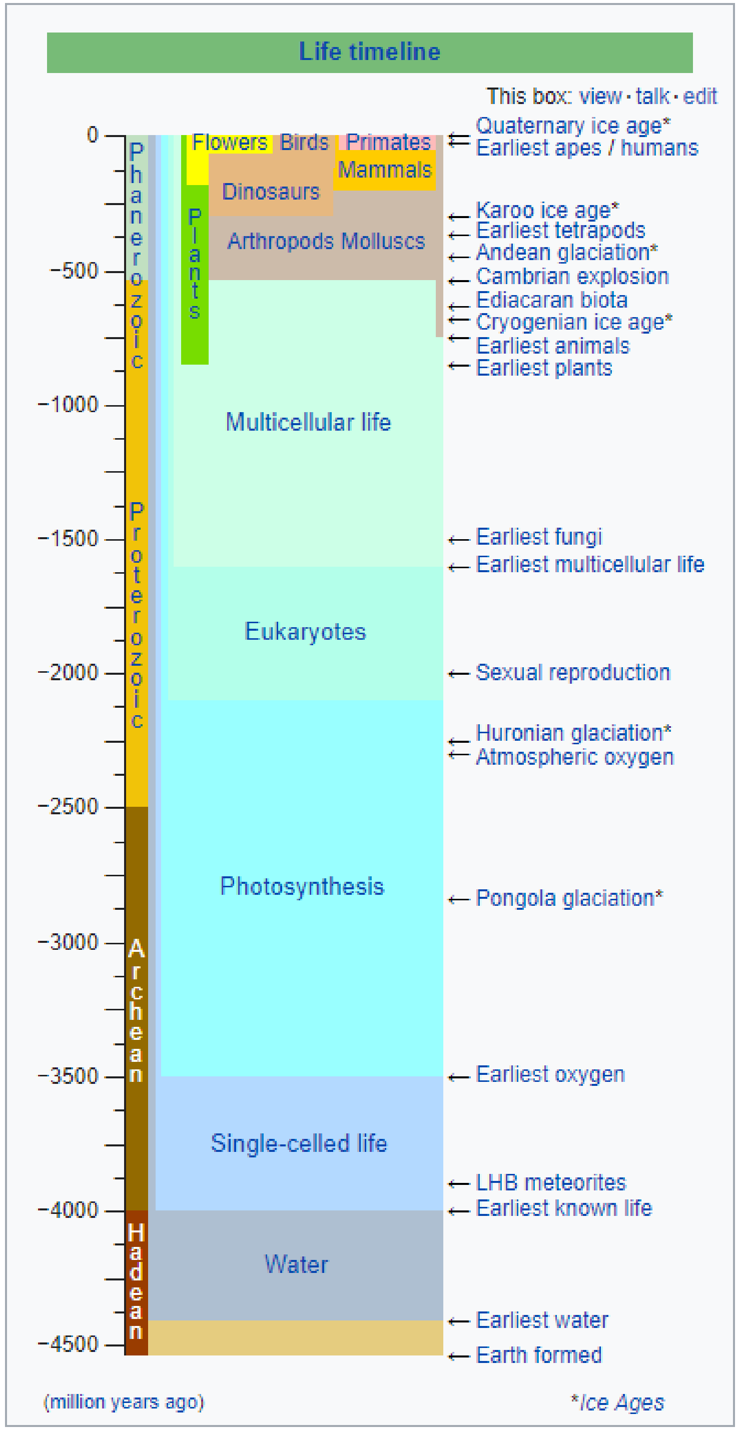
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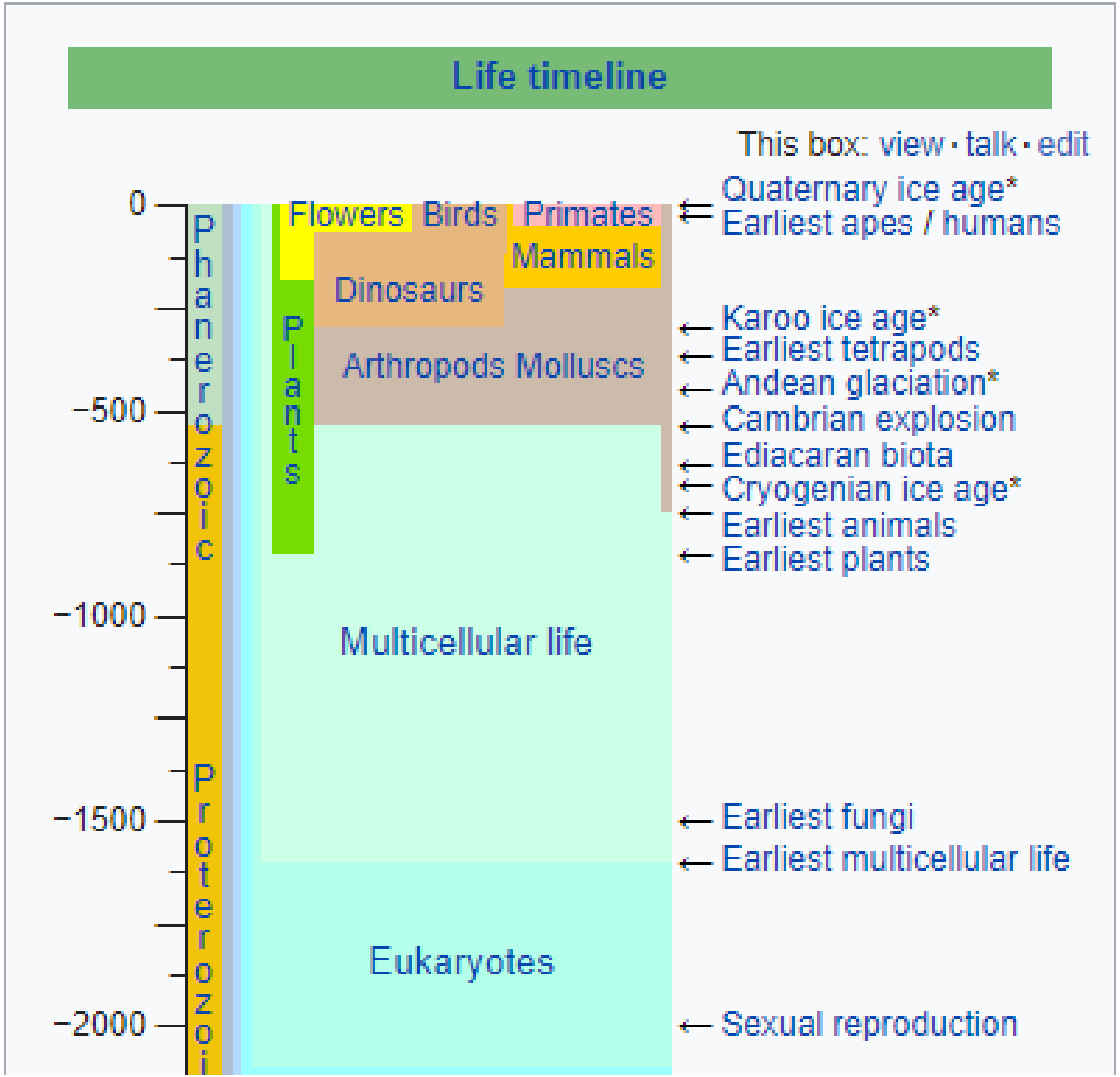
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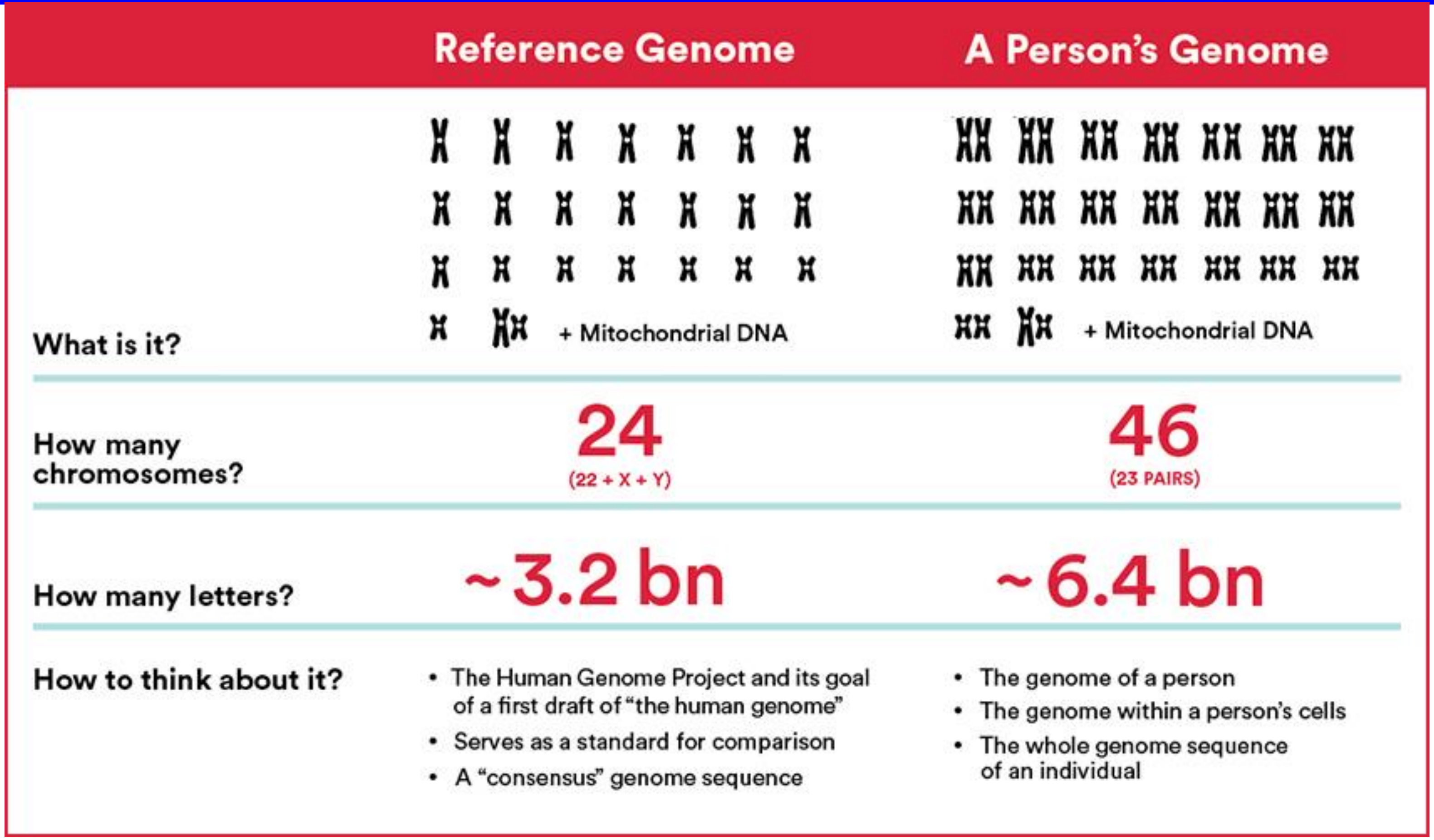
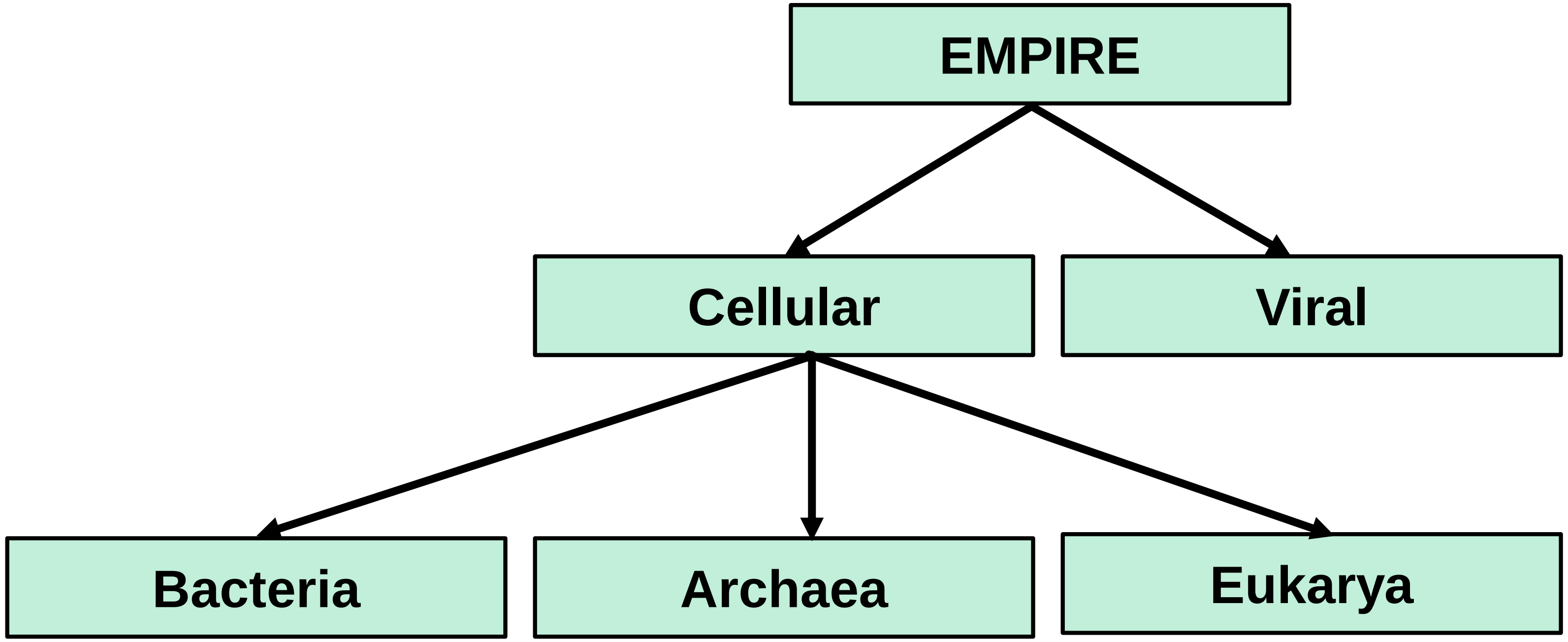
Sexual reproduction

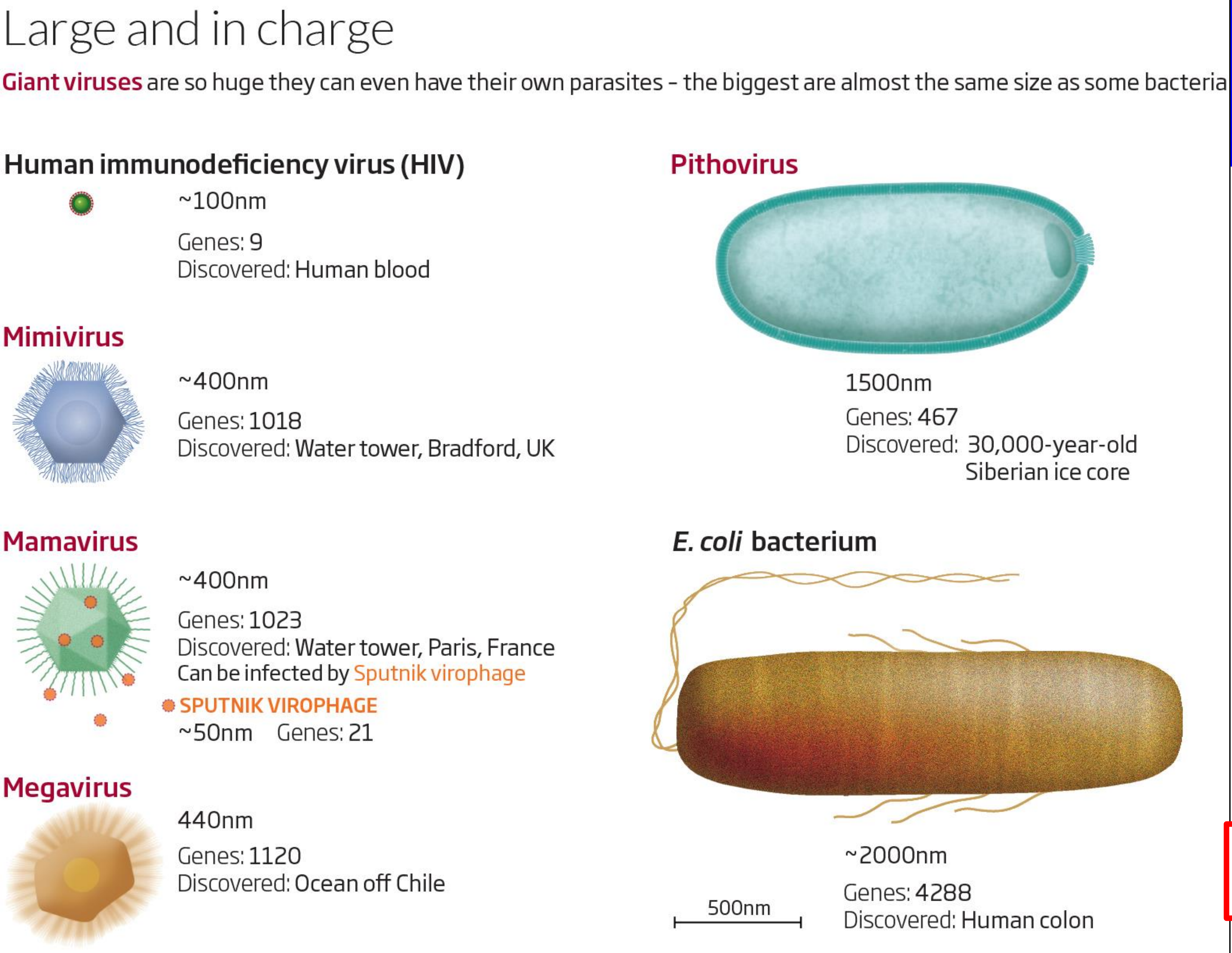
Human Genome



- That human genome–the real, physical human genome–is diploid; in other words, it has two pairs of each chromosome–one from each parent. Everyone has a chromosome 1 from mom and another from dad; and a chromosome 2 from mom and another from dad; and so forth. The total is 46 chromosomes, or two pairs of 23. Thus, 6.4 billion letters (or bases) long.
- The human genome contains about 3.2 billion base pairs of DNA. However, within the genome there are only about 20,000 genes that code for proteins, comprising only 1.5% of the genome.

Two empires





- ❑ Viruses are amazing molecular machines that are much tinier than even the smallest cells.
- ❑ They are intracellular parasites.
- ❑ These selfish genetic elements typically encode some proteins essential for viral replication, but they never contain the full set of genes for the proteins and RNAs required for translation, membrane function, or metabolism.
- ❑ Therefore, viruses exploit cells to produce their components.

Scientists identify more than 140,000 virus species in the human gut

Study opens up new research avenues for understanding how viruses living in the gut affect human health

Prof Rob Knight, from University of California San Diego, told the BBC: "You're more microbe than you are human."

Originally it was thought our cells were outnumbered 10 to one.

"That's been refined much closer to one-to-one, so the current estimate is you're about 43% human if you're counting up all the cells," he says.

But genetically we're even more outgunned.

The human genome - the full set of genetic instructions for a human being - is made up of 20,000 instructions called genes.

But add all the genes in our microbiome together and the figure comes out between two and 20 million microbial genes.

Prof Sarkis Mazmanian, a microbiologist from Caltech, argues: "We don't have just one genome, the genes of our microbiome present essentially a second genome which augment the activity of our own.

"What makes us human is, in my opinion, the combination of our own DNA, plus the DNA of our gut microbes."

“Eight percent of our DNA consists of remnants of ancient viruses, and another 40 percent is made up of repetitive strings of genetic letters that is also thought to have a viral origin.”

8 percent

Viral remnants make up 8 percent of the human genome, and a new study finds that these sequences are still active in healthy people.

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Reproduction

- Reproduction is a common characteristic of Living Things.
- In general terms, reproduction is defined as the biological process of giving birth to their young ones identical to their parents.

- There are two different types of reproduction:
  - **Sexual Reproduction**
  - When an organism reproduces sexually, it integrates genetic material from both parents to create a genetically unique organism.

  - **Asexual Reproduction** (Asexual reproduction occurs by binary fission, wherein a cell divides into two)
  - Binary fission is the most common mode of reproduction seen in bacteria.
  - Binary Fission: It is a common and simple method of asexual reproduction in bacteria.
  - In this process of asexual reproduction, the parent cell divides into two, leading to two equal-sized daughter cells. Under favourable conditions, bacteria multiply rapidly and may double in ~20 minutes.

Three domains of life

The tree of life consists of three domains: Bacteria, Archaea and Eukarya

- **All living organisms consist of elementary units called cells.**
- **All cells are believed to have evolved from a common progenitor because the structures and molecules in all cells have so many similarities.**
- **In recent years, detailed analysis of DNA sequences from a variety of Prokaryotic organisms revealed that they are of two distinct types: so-called "True" bacteria or Eubacteria and Archaea.**
- **Based on the assumption that the organisms with more similar genes evolved from a common progenitor more recently than those with more dissimilar genes. The researchers developed an Evolutionary Lineage Tree as shown in Figure 1.**
- **According to this tree, the Archaea and the Eukaryotes diverged from the Eubacteria and later they also diverged from each other.**

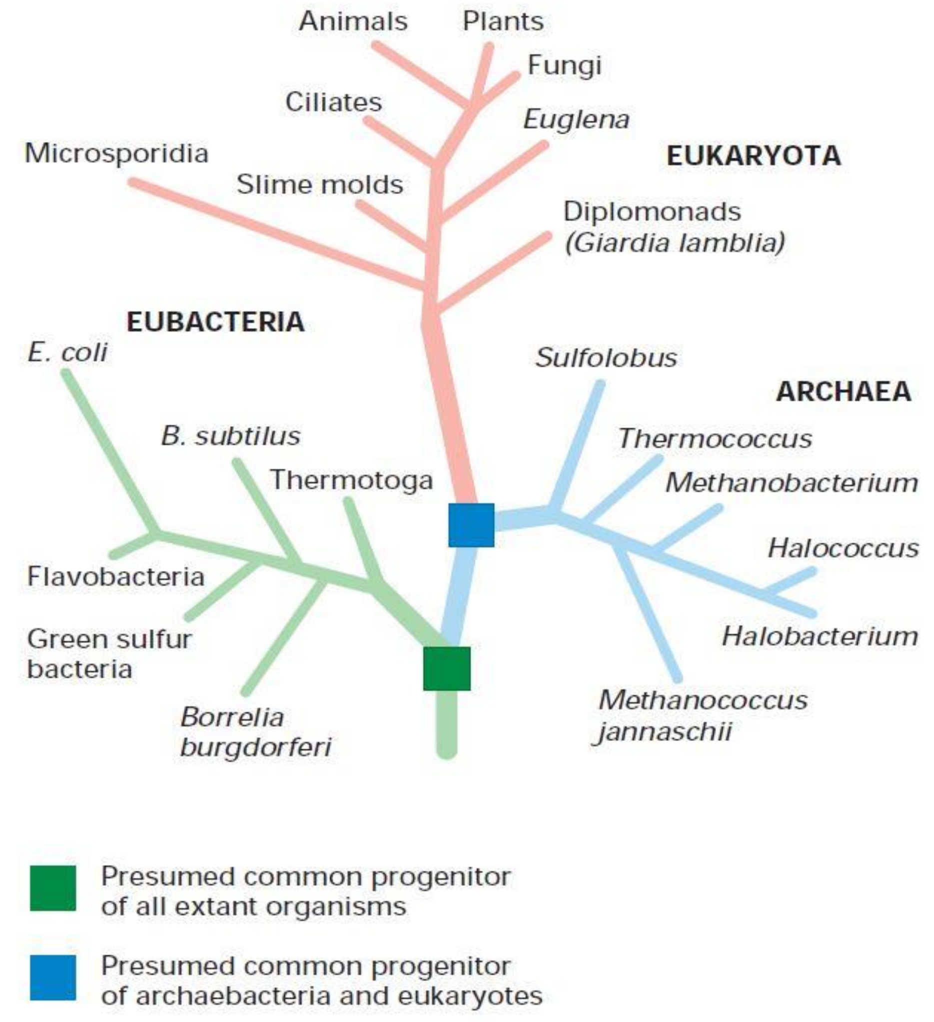
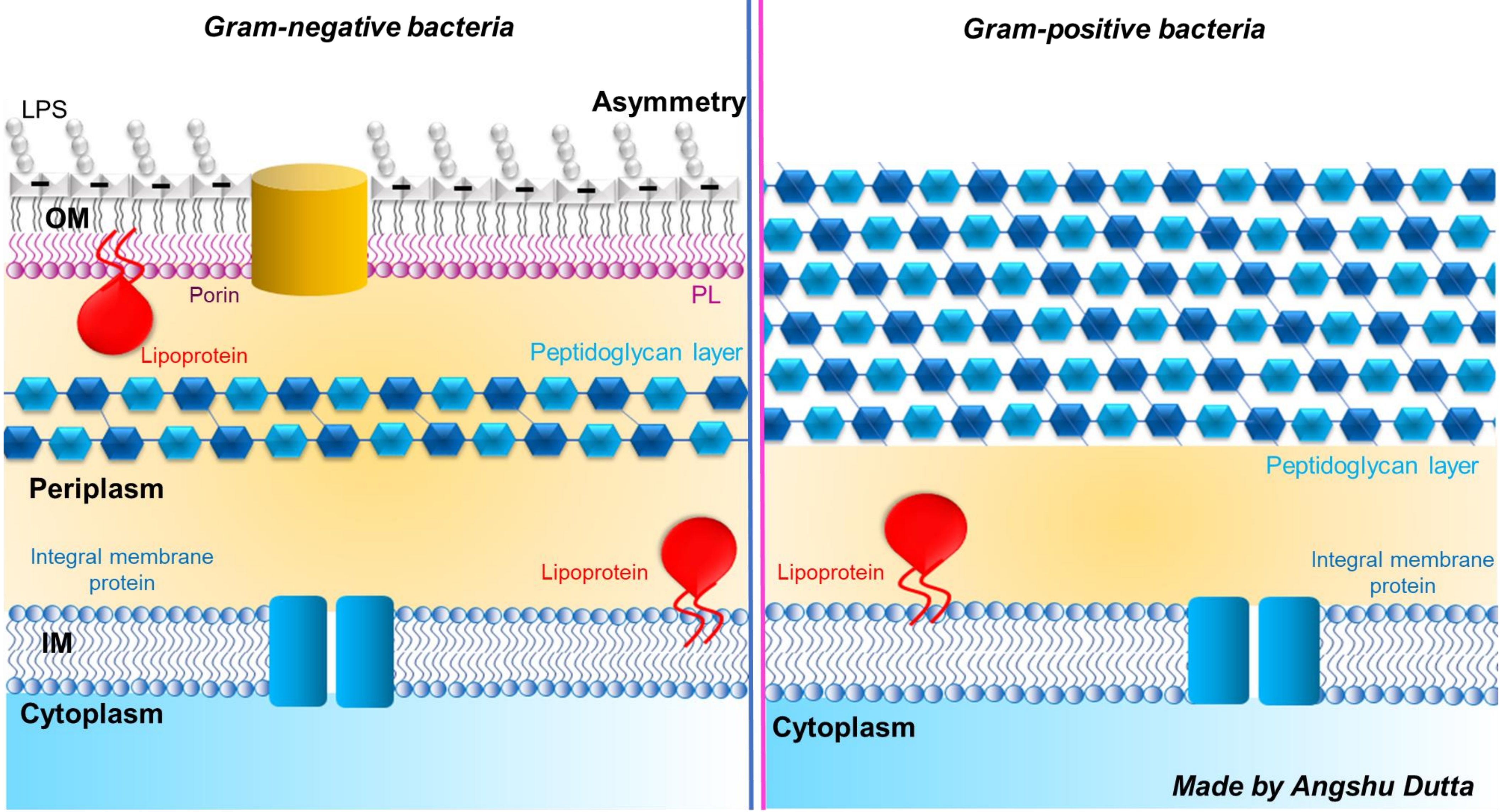
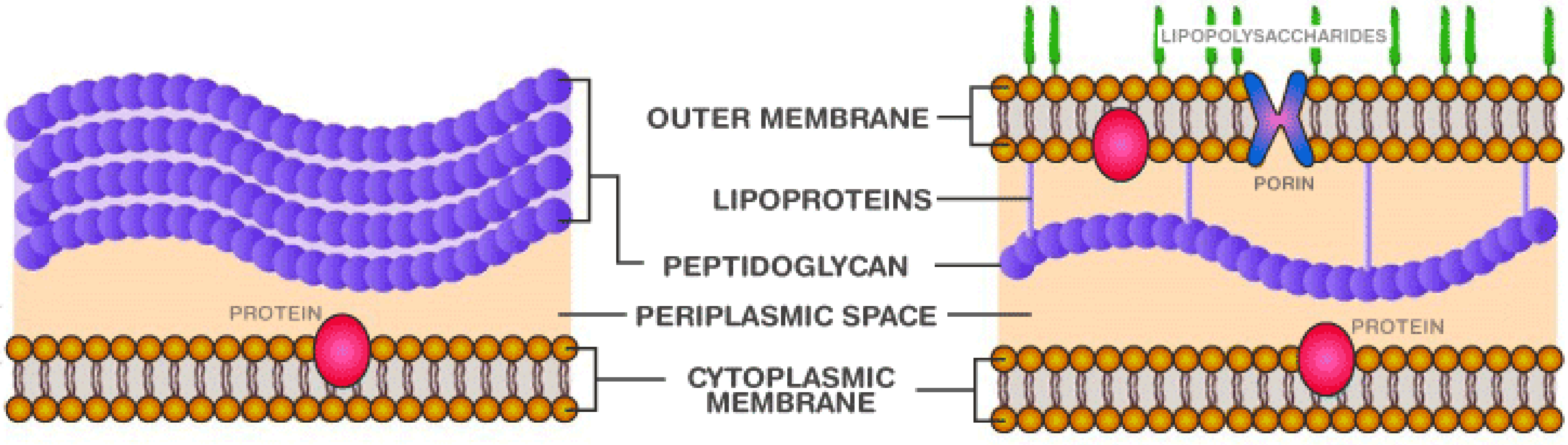


Fig. 1. All organisms from simple bacteria to complex mammals probably evolved from a common, single-celled progenitor. **Credit: Molecular Cell Biology - Lodish*

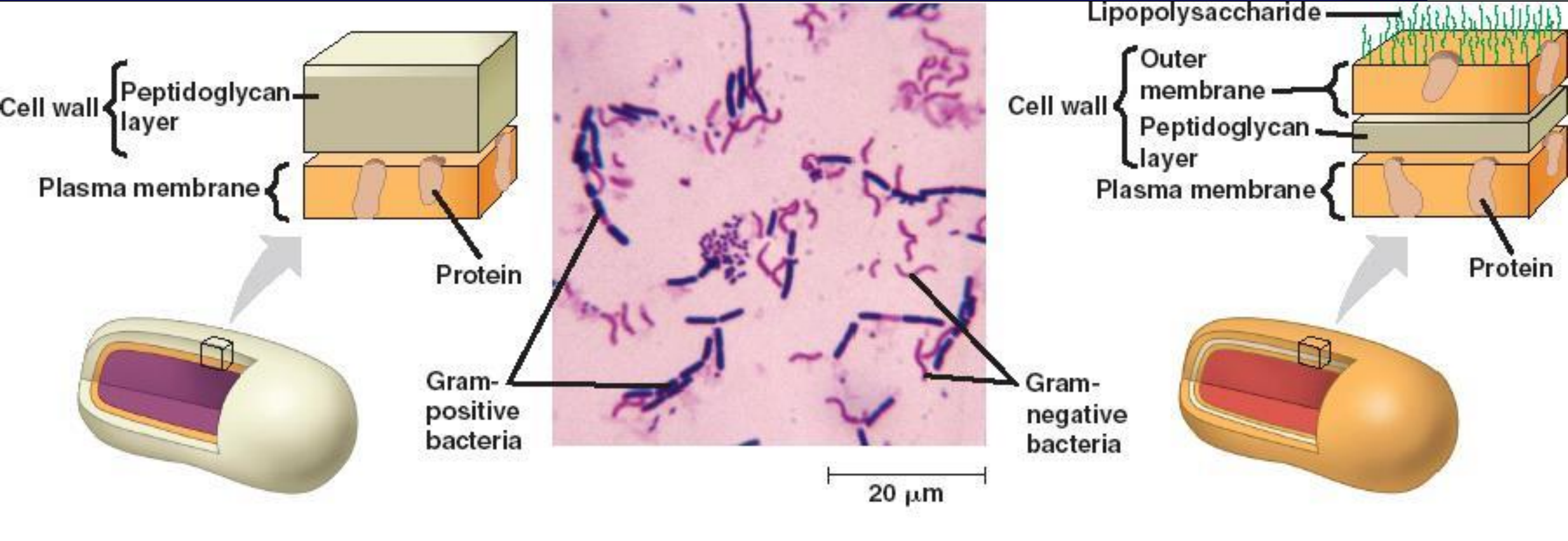


Gram-negative bacteria
Gram-positive bacteria
LPS
Asymmetry
OM
Porin
PL
Lipoprotein
Peptidoglycan layer
Periplasm
Integral membrane protein
Lipoprotein
IM
Cytoplasm
Peptidoglycan layer
Lipoprotein
Integral membrane protein
Cytoplasm
Made by Angshu Dutta



- Gram stain or Gram staining is a method of staining used to classify bacterial species into two large groups: **Gram-positive bacteria and Gram-negative bacteria.**
- Gram-positive bacteria have a thick mesh-like cell wall made of peptidoglycan (50–90% of cell envelope), and as a result are stained purple by crystal violet, whereas gram-negative bacteria have a thinner layer (10% of cell envelope), so do not retain the purple stain and are counter-stained pink by safranin.

Bacteria



(a) Gram-positive (b) Gram-negative

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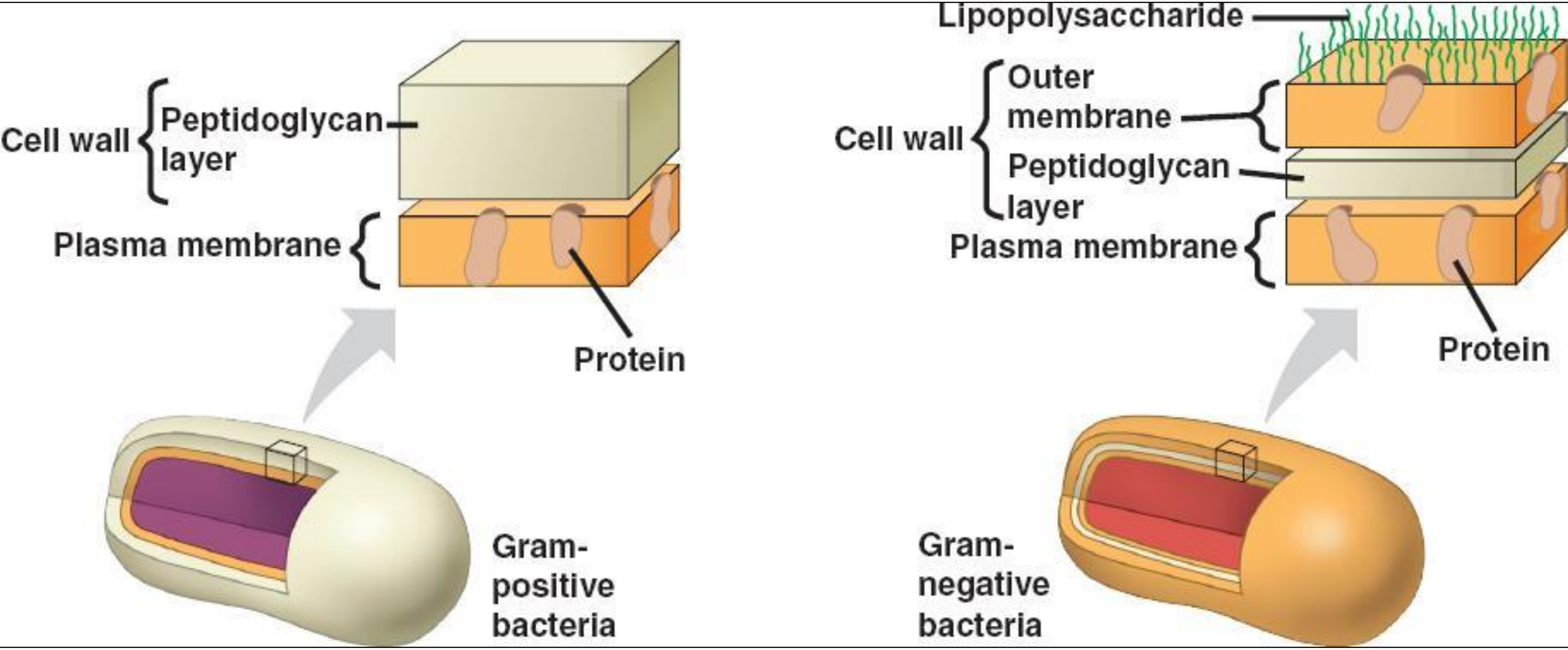
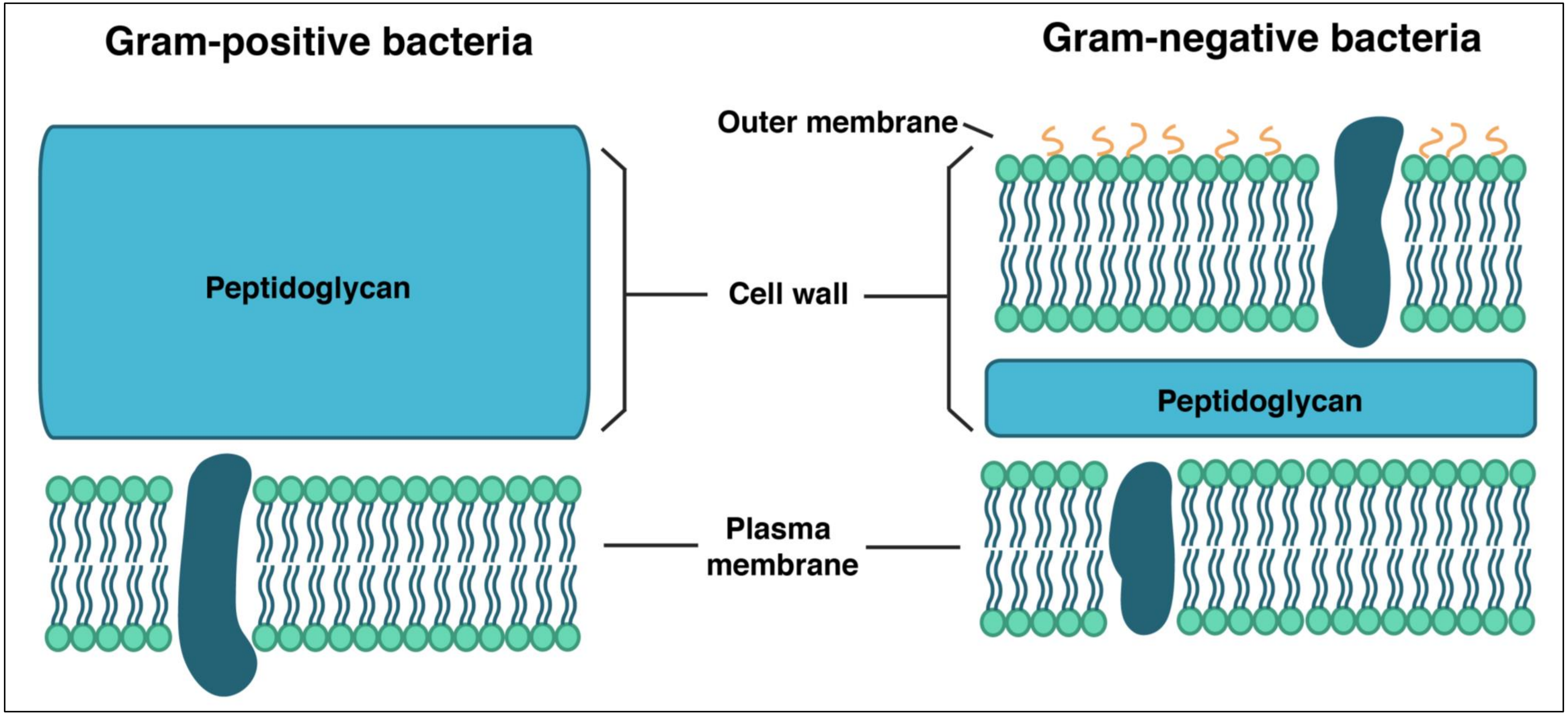


Cell wall
Peptidoglycan layer
Plasma membrane
Protein
Gram-positive bacteria
Lipopolysaccharide
Cell wall
Outer membrane
Peptidoglycan layer
Plasma membrane
Protein
Gram-negative bacteria

Bacteria



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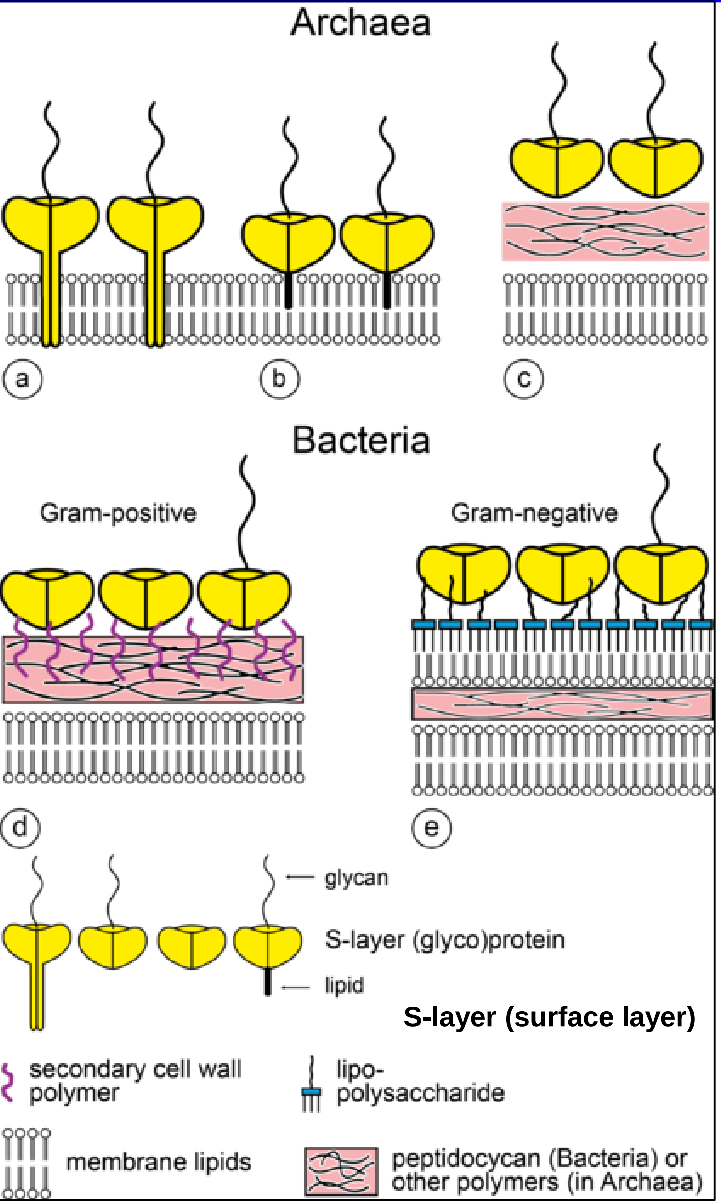
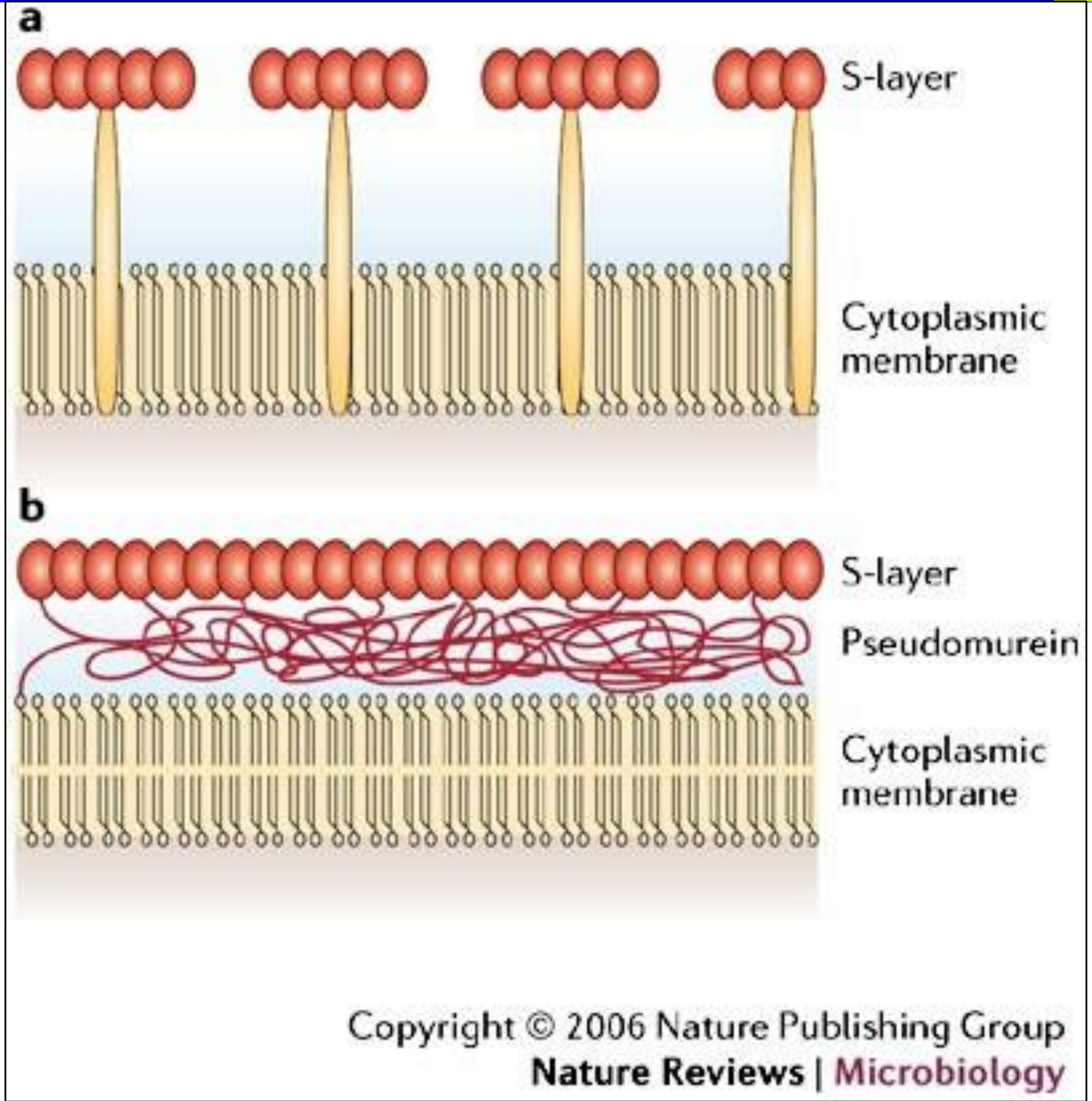
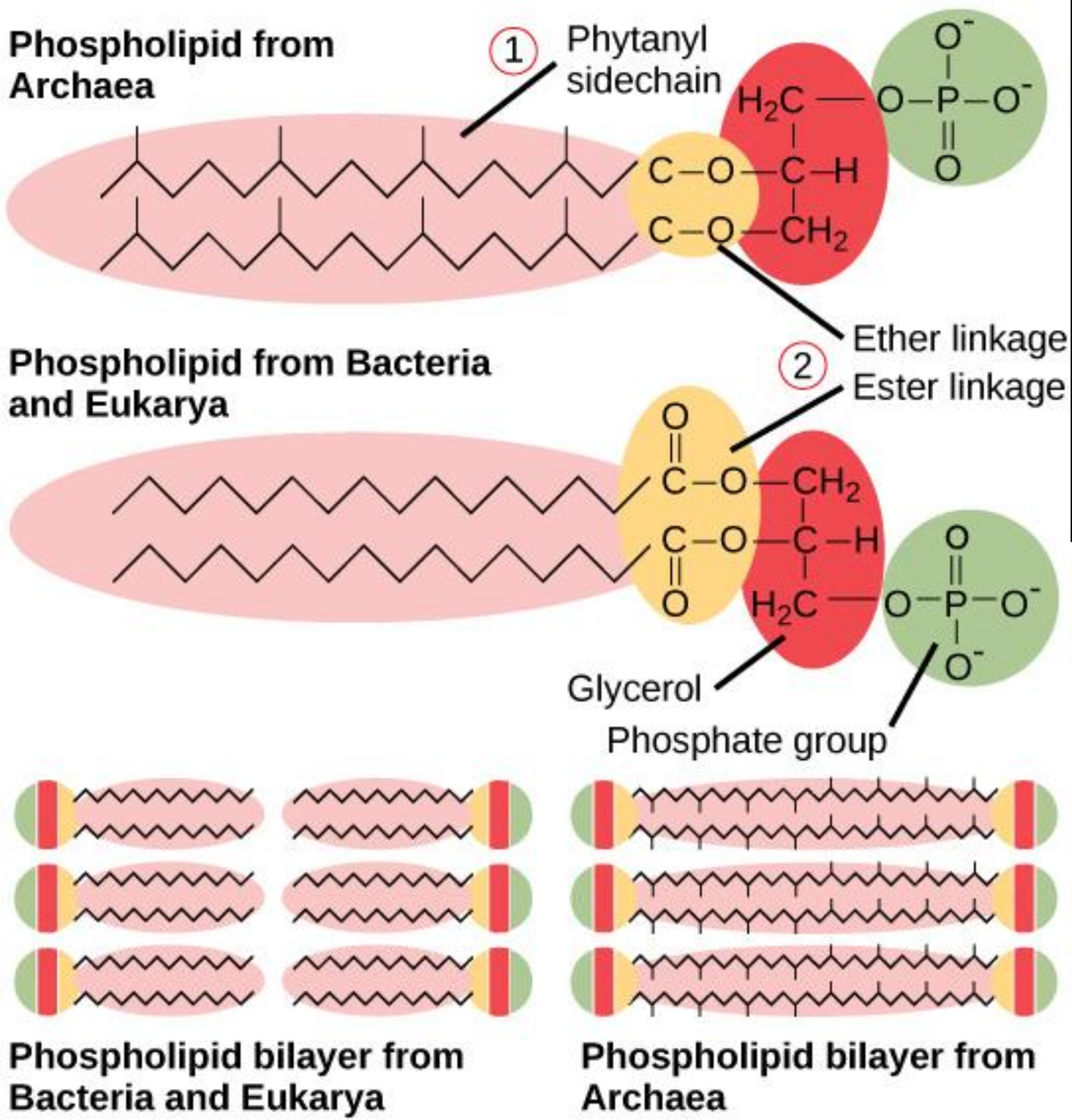
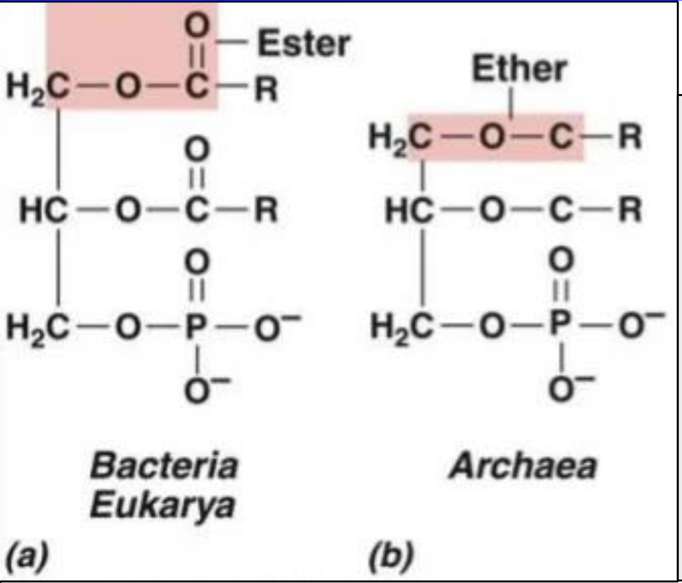
Archaea vs Eubacteria (cell wall)

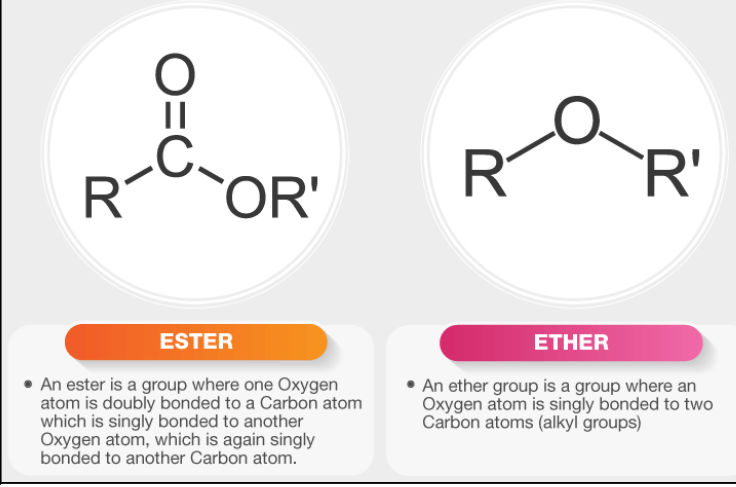




Archaea vs Eubacteria vs Eukarya (cell membrane)







- Archaeol is one of the main core membrane lipids of archaea, one of the three domains of life.
- Standard archaeol is 2,3-di-O-phytanyl-sn-glycerol, with two phytanyl chains binding to the position of sn-2 and sn-3 of glycerol by ether bonds.
- Archaeol is generally composed by linking two phytanyl chains to the sn-2 and sn-3 positions of a glycerol molecule. The highly branched side chains are speculated to account for the very low permeability of archaeol-based membrane, which may be one of the key adaptations of archaea to extreme environment.
- One of the key features that distinguishes archaea from bacteria and eukarya is their membrane lipids, where archaeol plays an important role.
- Because of this, archaeol is also broadly used as a biomarker for ancient archaea.

Important Links to see Videos

- **Bacteria:**
- https://www.youtube.com/watch?v=PJmtkQpqJxE
- https://www.youtube.com/watch?v=N7dTM7nLw1Q
- https://www.youtube.com/watch?v=pgr-HeVNbOY&t=391s

- **Three domains of life:**
- https://www.youtube.com/watch?v=6LZACcEmQtE

- **Archaea and the tree of life:**
- https://www.youtube.com/watch?v=hw-ij3822DY&list=PL6P3nr9Jc5Ioak1LId5QwCt6W0MIhGzQ9&t=435s
- https://www.youtube.com/watch?v=rd37jBXfM4k
- https://www.youtube.com/watch?v=7LdmcFoYQtU

- **Virus:**
- https://www.youtube.com/watch?v=YI3tsmFsrOg
- https://www.youtube.com/watch?v=FmX8au0xGlY&t=613s